

Roll No.

B T 2 3 0 0 2 0 C S

IT/W-25 / IT5M004 / DS5M004 / CS5M004



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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

AI5M004/01

B. Tech Summer / Winter - 2025
(ESE / ~~Make-up~~ / Supplementary) Examination

Branch: Information Technology/CSE(DS) | CSE / AI

Sem: V

Subject: Bias, Safety, Ethics in LLM

Subject Code: IT5M004/DS5M004 / CS / AI

Marks: 100

Date of Exam: 18/11/2025

Time: 3 Hrs. 5M004

Instructions:

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat diagrams.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

COs: Students should be able to:

1. Understand the key ethical principles and responsible AI frameworks in the context of large language models.
2. Examine the sources and consequences of bias in training data and outputs of LLMs, including gender, race, and cultural bias.
3. Analyze the mechanisms of hallucinations and misinformation in generative AI models and evaluate mitigation techniques.
4. Assess the modern AI auditing and de-biasing techniques and improve fairness in NLP and LLM systems.
5. Justify the model robustness, ethical vulnerabilities such as adversarial prompts, and propose content moderation strategies.

SECTION A

Q. No.	All Questions are compulsory. Answer in One/Two/Sentence.	Level / CO Mapped	Marks
Q.1	Define transparency in AI models.	L1 / CO1	2
Q.2	List two ethical concerns in NLP systems.	L1 / CO1	2
Q.3	What are racial and gender biases in data?	L1 / CO2	2
Q.4	State two consequences of biased datasets.	L2 / CO2	2
Q.5	Explain is AI hallucination	L2 / CO3	2
Q.6	List two sources of misinformation in LLM outputs.	L1 / CO3	2
Q.7	Describe the dataset auditing?	L2 / CO3	2
Q.8	Define adversarial input.	L1 / CO5	2
Q.9	Mention two methods to enhance model robustness.	L2 / CO5	2
Q.10	Discuss the significance of AI governance frameworks.	L3 / CO5	2
Q.11	List two AI fairness assessment tools.	L1 / CO4	2
Q.12	Define content filtering.	L1 / CO5	2

SECTION B

Q. No.	All Questions are compulsory.	Level / CO Mapped	Marks
Q.13	Explain the principle of fairness in Responsible AI.	L2 / CO1	4
Q.14	Describe three common biases in LLM datasets.	L3 / CO2	4
Q.15	Describe the types of hallucinations in generative AI systems.	L3 / CO3	4
Q.16	Discuss evaluation metrics used for fairness.	L3 / CO4	4
Q.17	Explain model moderation and filtering techniques.	L3 / CO5	4
Q.18	Outline the goals of global AI governance initiatives.	L4 / CO5	4

SECTION C

Q. No.	Solve any FOUR Question.	Level / CO Mapped	Marks
Q.19	Examine accountability as a pillar of ethical AI.	L4 / CO1	7
Q.20✓	Analyze the effects of bias in multilingual NLP models.	L4 / CO2	7
Q.21✓	Discuss mechanisms used to combat fake news using AI.	L4 / CO3	7
Q.22✓	Evaluate techniques for de-biasing model outputs.	L4 / CO4	7
Q.23✓	Explain prompt injection with ethical implications.	L5 / CO	7
Q.24	Summarize lessons learned from the ChatGPT case study.	L5 / CO5	7

SECTION D

Q. No.	Solve any TWO Question.	Level / CO Mapped	Marks
Q.25✓	Illustrate seven step decision making model with an example	L4/ CO1	12
Q.26✓	Propose a multi-step pipeline to ensure fairness in LLMs.	L6 / CO4	12
Q.27	Evaluate ethical and governance implications of global AI policies.	L6 / CO5	12